Mad Dog 5.4

Mad Dog 5.4 is a liquid herbicide that must be diluted with water according to label instructions before application. It is Glyphosate or Round UP. Mix thoroughly and apply to targeted weeds using a sprayer. Avoid drift to desirable plants and follow all safety precautions. Mad Dog 5.4 primarily targets broadleaf weeds and certain invasive plants.

SAFETY DATA SHEET

MAD DOG® 5.4

FOR CHEMICAL EMERGENCY, SPILL, LEAK, FIRE, EXPOSURE OR ACCIDENT, CALL CHEMTREC - DAY OR NIGHT 1-800-424-9300

1. IDENTIFICATION OF THE MATERIAL AND SUPPLIER

1.1 PRODUCT IDENTIFIER: **EPA REG. NO. 34704-929**
TRADE NAME: **MAD DOG® 5.4#**
1.2 RECOMMENDED USE: GROUP 9 HERBICIDE
1.3 SUPPLIER DETAILS:
LOVELAND PRODUCTS, INC.
P.O. Box 1286 • Greeley, CO 80632-1286
1.424 Hour Emergency Phone: 1-800-424-9300 - **Medical Emergencies:** 1-866-944-8565 – **Product Information:** 1-888-574-2878 (LPI-CUST)
U.S. Coast Guard National Response Center: 1-800-424-8802

2. HAZARDS IDENTIFICATION

2.1 Classification of the substance or mixture

Classification according to 29 CFR 1910.1200

Not classified

2.2 Label elements

None

Signalword:

HazardStatement: H303 – May be harmful if swallowed.
H333 – May be harmful if inhaled.

Precautionary Statement: (Prevention): P261 – Avoid breathing dust/fume/gas/mist/vapors/spray.
P262 – Do not get in eyes, on skin, or on clothing.
P264 – Wash thoroughly after handling.
P102 – Keep out of reach of children.

Precautionary Statement: (Response): P301+P330+P331: IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
P101 – If medical advice is needed, have the product container or label at hand.

Precautionary Statement: (Storage): P402+P234 – Store above 10 °F (-12 °C) to keep product from crystallizing. Keep only in original container.
P405 – Store locked up.

2.3 Other hazards

None known

3. COMPOSITION, INFORMATION ON INGREDIENTS

3.1 Substances

3.2 Mixtures

Classification according to 29 CFR 1910.1200

ChemicalName:	CASNo.	Concentration [%]
IsopropylaminesaltofGlyphosate	38641-94-0	53.80
*Otheringredients	n/a	46.20

*Ingredientsnotspecificallylistedarenon-hazardousoraretobeconsidered proprietary or confidentialbusiness information per 29 CFR 1910.1200(i)

4. FIRST AID MEASURES

4.1 Description of First Aid Measures

General Advice: Getmedical attention if symptoms occur.

Inhalation: Move person to fresh air. If person is not breathing, call 911 or an ambulance, then give artificial respiration, preferably by mouth-to-mouth if possible. Call a poison control center or doctor for further treatment advice.

Ingestion: Call a poison control center or doctor immediately for treatment advice. Have a person sip a glass of water if able to swallow. Do not induce vomiting unless told to do so by a poison control center or doctor. Do not give anything by mouth to an unconscious person. Call a poison control center or doctor for treatment advice.

Skin Contact: Take off contaminated clothing. Rinse skin immediately with plenty of water for 15-20 minutes. Call a poison control center or doctor for treatment advice.

Eye Contact: Hold eye open and rinse slowly and gently with water for 15-20 minutes. Remove contact lenses, if present, after the first 5 minutes then continue rinsing eye. Call a poison control center or doctor for treatment advice.

4.2 Most Important Symptoms and Effects, Acute and Delayed

Symptoms: Inhalation: May be harmful if inhaled.
Ingestion: May be harmful if swallowed.

4.3 ImmediateMedical Attention and Special Treatment

Treatment: Treatsymptomatically. Symptomsmaybe delayed.

FOR AMEDICAL EMERGENCYINVOLVINGTHISPRODUCTCALL : 1-866-944-8565

Take container, label or product name with you when seeking medical attention.

NOTES TO PHYSICIAN Treatment of exposure should be directed at the control of symptoms and the clinical condition of the patient.

5. FIRE FIGHTING MEASURES

5.1 EXTINGUISHING MEDIA:

Suitable ExtinguishingMedia: Drychemical, carbondioxide(CO_2),alcohol foam, foam, water spray or fog. Do not use water jet as thiswill spread thefire.

5.2 SPECIAL HAZARDS ARISING FROM THE SUBSTANCE OR MIXTURE:

Specific Hazards DuringFirefighting: During a fire, hazardous by-products can be released.

5.3 SPECIAL PROTECTIVE EQUIPMENT AND PRECAUTIONS FOR FIREFIGHTERS

Special Protective EquipmentforFirefighters:Self-contained breathing apparatus and full protective gear should be worn in fighting large fires involving chemicals. Use water spray to keep fire exposed containers cool. Keep people away. Isolate fire and deny unnecessary entry.

6. ACCIDENTAL RELEASE MEASURES

6.1 PERSONAL PRECAUTIONS, PROTECTIVE EQUIPMENT AND EMERGENCY PROCEDURES

Personal Precautions: Avoid inhalation of vaporsand spray mist and contact with skin and eyes. Ensure adequate ventilation. Wear suitable protective clothing.

6.2 ENVIRONMENTAL PRECAUTIONS

Environmental Precautions: Do not apply directly to water, or to areas where surface water is present or to intertidal areas below the meanhigh water mark. Do not contaminate water when disposing of equipment washwaters or rinsate.

6.3METHODS AND MATERIALS FOR CONTAINMENT AND CLEAN-UP

Methods for Clean-Up: Large Spills: Stop the flow of material, if this is without risk. Dike the spilled material, where this is possible. Absorb in vermiculite, dry sand or earth and place into containers. After removal flush contaminated area thoroughly with water.
Small Spills: Wipe up with absorbent material (e.g. cloth, fleece). Clean surface thoroughly to Remove residual contamination.
Never return spills to original containers for re-use.

7. HANDLING AND STORAGE

7.1 PRECAUTIONS FOR SAFE HANDLING:

AdviceonSafeHandling: Avoid inhalation of mists, vapors / spray and contact with eyes, skin and clothing. Do not breathe mists or vapor. Wear personal protective equipment. Do not use in areas without adequate ventilation. Avoid prolonged exposure. Wash thoroughly after handling. Do not empty into drains. Handle and open container with care. Use care in handling/storage. Wash before eating, drinking and/or smoking.

7.2 CONDITIONS FOR SAFE STORAGE:

RequirementsforStorageAreasandContainers: Store above 10 °F (-12.2 °C) to keep product from crystallizing. Crystals will settle to the bottom of the container. If allowed to crystallize, place in a warm room at 68 °F (20 °C) for several days to redissolve and roll or shake container or recirculate in mini-bulk or bulk containers to mix well before using. Do not contaminate water, foodstuffs, feed or seed by storage or disposal.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

8.1 CONTROL PARAMETERS:

OCCUPATIONAL EXPOSURE LIMITS

U.S. Workplace Exposure Level (ACGIH) TLVs

Components	Type	Value
No data available.		

U.S. Workplace Exposure Level (OSHA) PELs

Components	Type	Value
No data available.		

Biological limit values

ACGIH Biological Exposure Indices

Components	Value	Specimen
Nolistings		

8.2 EXPOSURE CONTROLS:

Engineering Measures

Whenhandlersuseclosedsystems, enclosedcabs, or aircraft in a manner that meets the requirements listed in Worker Protection Standard (WPS)foragricultural pesticides[40 CFR 170.240 (d) (4-6)], the handler PPE requirements may be reduced or modified as specified in the WPS.

Individual Protection Measures:

Goggles or shielded safety glasses are recommended.

Eye/FaceProtection: Long-sleeved shirt and long pants. Shoes plus socks.

SkinProtection: In case of inadequate ventilation or risk of inhalation of mists or vapors, use suitable respiratory equipment such as MSHA/NIOSH TC-21C or NIOSH approved respirator with N, R, P or HE filter. Wear respiratory protection during operations where spraying or misting occurs. If respirators are used, a program should be in place to assure compliance with 29 CFR 1910.134, the OSHA Respiratory Protection standard. Wear air supplied respiratory protection if exposure concentrations are unknown.

RespiratoryProtection:

9. PHYSICAL AND CHEMICAL PROPERTIES

9.1 APPEARANCE : Clear viscous liquid
ODOR: None.
ODOR THRESHOLD: No data available.
COLOR: Blue.
pH: 4.5 – 5.0 (1% solution)
MELTING POINT / FREEZING POINT: No data available
BOILING POINT: No data available
FLASH POINT: >200 °F / >93.3 °C.
FLAMMABILILITY (solid, gas): No data available.
UPPER / LOWER FLAMMABILITY OR EXPLOSIVE LIMITS: No data available.
VAPOR PRESSURE: No data available.
SOLUBILITY: Miscible
PARTITION CO-EFFICIENT, n-OCTANOL / WATER: No data available.
AUTO-IGNITION TEMPERATURE: No data available.
DECOMPOSITION TEMPERATURE: No data available
VISCOSITY: (kinematic): No data available
SPECIFIC GRAVITY (Water = 1): 1.205 – 1.227 g/ml
DENSITY: 10.06 – 10.24 lbs./gal / 1.21 – 1.23 kg/L

Note: These physical data are typical values based on material tested but may vary from sample to sample.
Typical values should not be construed as a guaranteed analysis of any specific lot or as specification items.

10. STABILITY AND REACTIVITY

10.1 REACTIVITY
Stable
10.2 CHEMICAL STABILITY
Stable undernormal temperatureconditions
10.3 POSSIBILITY OF HAZARDOUS REACTIONS
No data available. Willnotpolymerize.
10.4 CONDITIONS TO AVOID
Use of galvanized or unlinedsteel.
10.5 INCOMPATIBILE MATERIALS
This product and its spraysolutionswillreact with galvanized or unlined steel to produce hydrogen gas that may form a highly combustible gas mixture, which could flash or explode if ignited.Acids and bases.
10.6 HAZARDOUS DECOMPOSITION PRODUCTS
Will emit toxic fumes as itburns.

11 TOXICOLOGICAL INFORMATION

11.3 LIKELY ROUTES OF EXPOSURE
Inhalation. Ingestion.
LC50 (rat)>2.09mg/L(4HR)(Data from a similar product)
LD50 Oral (rat):>5,000mg/kg
LD50 Dermal (rat):>5,000mg/kg
Acute Toxicity Estimates: No data available
Skin Irritation (rabbit): Slight irritant.
Eye Irritation (rabbit): Minimal irritant
Specific Target Organ Toxicity: Single exposure: No data available.
Aspiration: Nodataavailable
Skin Sensitization (guinea pig): Not a sensitizer
Carcinogenicity: Nodataavailable
Germ Cell Mutagenicity: No data available
Interactive Effects: Noneknown

12 ECOLOGICAL INFORMATION

12.3 ECOTOXICITY

Theproductisnotclassified as environmentally hazardous. However, this does not exclude the possibility that large or frequent spills can have a harmfulordamaging effect on the environment.

Ecotoxicological Data

	Species	Test Results
Glyphosate acid	Rainbow trout	5.4 mg/L – 96-hour LC50
	Bluegill	7.3 mg/L – 96-hour LC50
	Bee	>100µ/bee – Contact LD50

Drift or runoff may adversely affect non-target plants.

Do not apply directly to water.
Do not contaminate water when disposing of equipment wash water.
Do not apply when weather conditions favor drift from target area.

12.2PERSISTENCE AND DEGRADABILITY

Biodegradability: No data available

12.3BIOACCUMULATIVE POTENTIAL

Bioaccumulation: No data available.

12.4MOBILITY IN SOIL

No data available.

12.5OTHER ADVERSE EFFECTS

Assessment: No data available.

13 DISPOSAL CONSIDERATIONS

13.1 WASTE TREATMENT METHODS

PESTICIDE DISPOSAL: Wastesresultingfromtheuseofthisproductthatcannotbeusedorchemicallyreprocessedmustbedisposedofinalandfill approvedforpesticidedisposalorinaccordancewithapplicablefederal,state,orlocalprocedures.Emptiedcontainerretainsvaporandproductresidue. Observe all labeled safeguards until containeris cleansed, reconditioned, or destroyed.

CONTAINER HANDLING: Nonrefillable container. Donotreusethiscontainertoholdmaterialsotherthanpesticidesordilutepesticides(rinsate). Afteremptyingandcleaning,itmaybeallowabletotemporarilyholdrinsateorotherpesticide-relatedmaterialsinthecontainer.Contactyourstate regulatoryagencytodetermineallowablepracticesinyourstate.Oncecleaned,someagriculturalplasticpesticidecontainerscanbetakentoa containercollectionsiteorpickedupforrecycling.Tofindthenearestsite,contactyourchemicaldealerormanufacturer,orcontactTheAgricultural Container Recycling Council (ACRC) at www.acrecycle.org. Triple rinse or pressure rinse container (or equivalent) promptlyafter emptying. **For packagesuptoandequalto5gallons:Triplerinseasfollows:**Emptytheremainingcontentsintoapplicationequipmentoramixtankanddrain for 10 secondsaftertheflowbeginstodrip.Fillthecontainer1/4fullwithwaterandrecap.Shakefor10seconds.Pourrinsateintoapplication equipmentoramixtankorstorerinsateforlateruseordisposal.Drainfor10secondsaftertheflowbeginstodrip.Repeatthisproceduretwomore times. **Pressurerinseasfollows:**Emptytheremainingcontentsintoapplicationequipmentoramixtankandcontinuetodrainfor10secondsafterthe flow beginstodrip.Holdcontainerupsidedownoverapplicationequipmentormixtankorcollectrinsateforlateruseordisposal.Insertpressurerinsing nozzle inthesideofthecontainer,andrinseatabout40PSIforatleast30seconds.Drainfor10secondsaftertheflowbeginstodrip.**Forpackages greaterthan5gallonsandlessthan56gallons:Triplerinseasfollows:**Emptytheremainingcontentsintoapplicationequipmentoramixtank.Fill the container1/4fullwithwater.Replaceandtightenclosures.Tipcontaineronitssideandrollitbackandforth,ensuringatleastonecomplete revolution,for30seconds.Standthecontaineronitsendandtipitbackandforthseveraltimes.Emptytherinsateintoapplicationequipmentoramix tank orstorerinsateforlateruseordisposal.Repeatthisproceduretwomoretimes.

Pressure rinse as follows: Emptytheremainingcontentsinto applicationequipmentoramixtankandcontinuetodrainfor10secondsaftertheflowbeginstodrip.Holdcontainerupsidedownoverapplication equipmentormixtankorcollectrinsateforlateruseordisposal.Insertpressurerinsingnozzleinthesideofthecontainer,andrinseatabout40PSIfor at least30seconds.Drainfor10secondsaftertheflowbeginstodrip.

For packages greater than 56 gallons: Tocleanthecontainerbeforefinal disposal,emptytheremainingcontentsfromthiscontainerintoapplicationequipmentormixtank.Fillthecontainerabout10percentfullwithwater. Agitatevigorouslyorrecirculatewaterwiththepumpfor2minutes.Pourorpumprinsateintoapplicationequipmentorrinsatecollectionsystem.Repeat this rinsingproceduretwomoretimes. **For refillable containers:** Refillthiscontainerwithpesticideonly.Donotreusethiscontainerforanyother purpose.Cleaningthecontainerbeforefinaldisposalistheresponsibilityofthepersondisposingofthecontainer.Cleaningbeforerefillingisthe responsibilityoftherefiller.Tocleanthecontainerbeforefinaldisposal,emptytheremainingcontentsfromthiscontainerintoapplicationequipmentor mix tank.Fillthecontainerabout10percentfullwithwater.Agitatevigorouslyorrecirculatewaterwiththepumpfor2minutes.Pourorpumprinsateinto applicationequipmentorrinsatecollectionsystem.Repeatthisrinsingproceduretwomoretimes.Donotcontaminatewater,foodorfeedbystorageor disposal.

14 TRANSPORT INFORMATION

14.3 LAND TRANSPORT
DOT Shipping Description: NOTREGULATED BY USDOT
U.S. Surface Freight Classification: COMPOUND, TREE OR WEED KILLING, NOI (NMFC 50320, SUB 2: CLASS 60)

15 REGULATORY INFORMATION

15.1 SAFETY, HEALTH AND ENVIRONMENTAL REGULATIONS

NFPA & HMIS Hazard Ratings:

NFPA		HMIS	
1	Health	1	Health
0	Flammability	0	Flammability
1	Instability	1	Reactivity
		C	PPE

0	Least
1	Slight
2	Moderate
3	High
4	Severe

SARA Hazard Notification/Reporting
SARA Title III Hazard Category:

Immediate	Y	Fire	N	Sudden Release of Pressure	N
Delayed	N	Reactive	N		

Reportable Quantity (RQ) under U.S. CERCLA: Not listed.
SARA, Title III, Section 313: Not listed.
RCRA Waste Code: Not listed.
CA Proposition 65: Not applicable

This chemical is a pesticide product registered by the United States Environmental Protection Agency and is subject to certain labeling requirements under federal pesticide law. These requirements differ from the classification criteria and hazard information required for safety data sheets (SDS), and for workplace labels of non-pesticide chemicals. Following is the hazard information as required on the pesticide label:

CAUTION

Domestic Animals: This product is considered to be relatively nontoxic to dogs and other domestic animals; however, ingestion of this product or large amounts of freshly sprayed vegetation may result in temporary gastrointestinal irritation (vomiting, diarrhea, colic, etc.). If such symptoms are observed, provide the animal with plenty of fluids to prevent dehydration. Call a veterinarian if symptoms persist for more than 24 hours.

16 OTHER INFORMATION

SDS STATUS: Sections1,2, 13,and15 revised.
PREPARED BY: Registrations and Regulatory Affairs
REVIEWED BY: Environmental Health and Safety

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EPA REG. NO.: 34704-929

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